

Microsoft and Google's staggering water consumption rates for AI

Posted by u/Alsupercharged • 0 points • 22 comments

AI, with its vast resource needs, is raising concerns over sustainability and environmental impact. Last year, Microsoft's data centers drained over 2,500 Olympic-sized swimming pools worth of water, reflecting a 34% increase from the previous year. Google also reported a 20% water consumption increase over the same period.

To stay on top of the latest advancements in AI, [look here first.](https://supercharged-ai.beehiiv.com/subscribe?utm_source=reddit&utm_medium=ai-water&utm_campaign=campaign)

<https://preview.redd.it/47vf3fi1crnb1.png?width=990&format=png&auto=webp&s=ba7f29496e979e066777eb390c2cb645046b7e6d>

****A deeper look at AI's water footprint****

- * The growth of AI and related technologies increases the need for vast server farms, which depend heavily on water for cooling purposes.
- * The spike in water usage can be attributed primarily to AI, as per Shaolei Ren, a researcher at the UC, Riverside, who focuses on AI's environmental impact.
- * For every 5 to 50 prompts submitted to ChatGPT, it consumes about 500 ml of water, according to an upcoming paper from Professor Ren's team.

****Big Tech and Responsible Water Usage****

- * Recognizing their significant water consumption, tech companies like Google have voiced concerns and are exploring ways to mitigate the negative effects.
- * Google has committed to responsible water usage, which includes assessing where and how their water usage might affect surrounding areas.

[(source)](<https://www.independent.co.uk/tech/ai-water-energy-artificial-intelligence-b2409356.html>)

****P.S.** If you like this kind of analysis, **** I write [a free newsletter](https://supercharged-ai.beehiiv.com/subscribe?utm_source=reddit&utm_medium=ai-water&utm_campaign=campaign)** that explores the latest AI developments. Professionals

from Google, Meta, and OpenAI are already reading it.

Comments

u/datascience-ModTeam • 1 points • 2023-10-03 16:25 UTC

I removed your submission. We prefer to minimize the amount of promotional material in the subreddit, whether it is a company selling a product/services or a user trying to sell themselves.

Thanks.

u/scott_steiner_phd • 1 points • 2023-09-13 02:48 UTC

Why post the Independant's hot take when you could post the [AP's original reporting] (<https://apnews.com/article/chatgpt-gpt4-iowa-ai-water-consumption-microsoft-f551fde98083d17a7e8d904f8be822c4>) instead?

u/pythonwiz • 1 points • 2023-09-12 21:40 UTC

Probably losses due to permeability of tubes and the sheer size of the system. Even closed loop systems need to be topped off from time to time. It could also be the use of evaporative cooling for datacenter AC which is very energy efficient in the right conditions, but uses water.

u/dxhunter3 • 3 points • 2023-09-12 11:47 UTC

Is there a research article or report related to this with methods and references/data sources? It is interesting but concerning. I work in the water industry and with climate so I am interested but with out the methodology it is just propaganda.

u/FlatRobots • 2 points • 2023-09-12 10:01 UTC

What do they do with all the water? Where does it go?

u/Sphagnum_Shuffle • 2 points • 2023-09-12 10:06 UTC

I assume they use the water for cooling the servers

u/FlatRobots • 2 points • 2023-09-12 10:15 UTC

So the water isn't really used up or contaminated? It goes back into nature, just a few degrees warmer.

u/Sphagnum_Shuffle • -2 points • 2023-09-12 10:21 UTC

I guess some of the water evaporates during the cooling process. And I think it would be quite bad for the local ecosystem if the water temperature increases few degrees

u/FlatRobots • 6 points • 2023-09-12 10:29 UTC

Warming up rivers is certainly a problem around industrial facilities, nuclear power plants etc and I don't want to downplay that. But whenever I read articles about how much water certain industries, crops etc "use", I'm always sceptical what "use" really means.

Most of time, the water just goes back into the environment, it's not gone or contaminated after being used. Many articles make it sound like the water is gone, like energy.

u/soxfan15203 • 15 points • 2023-09-12 09:01 UTC

Much like nuclear power plants, can't the water just be pumped back into the environment? If it's only used for cooling purposes, it shouldn't be contaminated, right?

u/scott_steiner_phd • 2 points • 2023-09-13 02:54 UTC

> Much like nuclear power plants, can't the water just be pumped back into the environment? If it's only used for cooling purposes, it shouldn't be contaminated, right?

Into the environment, probably eventually, but not necessarily when it's needed, or into the water supplies from what it's drawn. From the [original article](<https://apnews.com/article/chatgpt-gpt4-iowa-ai-water-consumption-microsoft-f551fde98083d17a7e8d904f8be822c4>) the Independent (and OP) are hijacking,

> For much of the year, Iowa's weather is cool enough for Microsoft to use outside air to keep the supercomputer running properly and vent heat out of the building. Only when the temperature exceeds 29.3 degrees Celsius (about 85 degrees Fahrenheit) does it withdraw water, the company has said in a public disclosure.

> That can still be a lot of water, especially in the summer. In July 2022, the month before OpenAI says it completed its training of GPT-4, Microsoft pumped in about 11.5 million gallons of water to its cluster of Iowa data centers, according to the West Des Moines Water Works. That amounted to about 6% of all the water used in the district, which also supplies drinking water to the city's residents.

> In 2022, a document from the West Des Moines Water Works said it and the city government "will only consider future data center projects" from Microsoft if those projects can "demonstrate and implement technology to significantly reduce peak water usage from the current levels" to preserve the water supply for residential and other commercial needs.

Pulling increasing amounts of water out of a municipal water supply during peak water use season and either returning it in fall or venting it into the atmosphere isn't necessarily sustainable just because the H₂O molecules aren't destroyed, for the same reasons that pulling increasing amounts of water out of the Colorado River in the summer and pouring it into your lawn isn't sustainable.

u/Ok_Unit7169 • 1 points • 2023-09-13 06:28 UTC

Why not build data centers like northern Wisconsin, where temperatures can even go below 50 F? That way, they can use outside ambient temperature.

u/slowpush • 1 points • 2023-09-13 12:25 UTC

Skill shortages are in those areas.

u/bobby_table5 • 16 points • 2023-09-12 09:48 UTC

It is: those are closed systems that warm the air or another warm circuit. That's why the article feels strange to me. Nuclear power plants have some of that water leaving (you can see the white plumes from the big chimneys), but it's a tiny fraction and only because you can't safely

guarantee there's enough cooling without it. Data centers emit a lot less heat and do not have a risk of emitting a lot more heat than expected.

If they do, there are simple solutions: I'd be all for shaming them into doing the right thing, but it doesn't match the internal culture of those teams.

u/Clear-Can6701 • 12 points • 2023-09-12 07:57 UTC

So can someone more intelligent enlighten me, how this water "consumption" is an issue?

In my mind, that water is still water, even after it has absorbed heat and can still be used for whatever things water is used for. It's not like these data centers contaminate the water like industries like the textile industry would.

u/bobby_table5 • 2 points • 2023-09-12 09:45 UTC

I'm very surprised by those articles because I'm familiar with efforts to convert data centers to more sustainable practices, and you typically either go

1. from
 1. using a closed water loop that heats up the air with big vents;
 2. using a closed water loop to warm a block/city-wide hot water network
2. or from
 1. using open-air cooling
 2. to add a closed water loop to warm 1. a block/city-wide hot water network

Neither of those just boil water off, and that's it. It would be an efficient and compact design if you are space-constrained and have plenty of water, but I've never heard of anything thinking that's a good idea.

Nuclear power plants and large plants (those that use the giant concrete paraboloid towers that people associate with nuclear plants) do use an open water cycle to cool things (vapor goes up the tower, precipitates, and rains back inside the concrete structure: they are there to catch the water back) but those are meant to waste as little water as possible, and cool things that are a lot, lot warmer than a data center.

u/Sir_-Butters22 • 65 points • 2023-09-12 07:45 UTC

I don't see this as a problem, and it's difficult to gauge why there was an increase, and pointing at AI when Cloud Data Centers are the backbone of the majority of the internet now.

Also, for reference:

An average person's yearly usage of water is 100,000 litres

An Olympic Swimming Pool is 2,500,000 liters

So 25 people account for one Olympic swimming pool

So the 25,000 Olympic Swimming Pools quotes accounts for 62,500 people.

If these are global figures, this is some bullshit article trying to shine AI progress as an environmental problem...

u/foxbatcs • 2 points • 2023-09-12 23:32 UTC

I always see red flags whenever someone tries to use a non-standard unit for something like this.

u/CaptnSe13 • 1 points • 2023-09-12 20:42 UTC

While not changing anything to the conclusion, 25 people x 25,000 pools = 625,000 people, rather than 62,500 people.

But again, not changing anything to the conclusion.

u/Sir_-Butters22 • 1 points • 2023-09-12 20:50 UTC

Oh, yeah my bad, thank you for pointing it out.

u/drunkenlazersharks • 2 points • 2023-09-12 20:27 UTC

That's just Microsoft though, and a mid-sized city's worth of water. Definitely not a global figure.

u/Sir_-Butters22 • 17 points • 2023-09-12 07:48 UTC

Additionally, I would like to see how much water/energy usage will be used if everybody used on-prem data centers, the efficiency gains of centralised compute/data centers are insane.

u/Binliner42 • -4 points • 2023-09-12 07:16 UTC

Refreshing post.